Malik Nauman Rauf

CONTACT INFORMATION

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RESEARCH INTERESTS

My research interests include flight physics, control systems and engineering, with a focus on model predictive control and adaptive control, as well as optimal control. I am also deeply interested in the modeling of uncertain and partially controllable systems, stochastic systems, astrodynamics, and astrophysics.

EDUCATION

University of Illinois, Urbana-Champaign, IL US

Ph.D Mechanical/Aerospace Engineering, MS in Mathematics, August 2026 - *Present*

- Working on Problems of Transfomer based optimal control and estimator design.
- Courses: ECE 586GT, ECE 534
- Course Projects:
  - "Multi-UAV Cooperative Search & Spatial Coverage; A Game-Theoretic development"; Python; ECE 586GT, Game Theory

Georgia Institute of Technology, Atlanta, Georgia US

M.S. Aerospace Engineering, August 2024 - May 2026

- Grade: 4.00/4.00
- Fulbright Scholar
- Primary Area: Flight Mechanics and Control
- Courses: AE 6530, MATH 6579, AE 8803 VAM, ECE 6553, MATH 6338, AE 6580, ECE 6271, MATH 6307, MATH 7338, ECE 8903, AE 6353, AE 6531
- Course Projects:
  - "Implementation of Q-learning for Multi-Agent Adversarial Games"; Python; AE 8803 Optimal Control for Learning and Games
  - "Safety Promoting Nearly-Optimal Neural Lyapunov Controllers"; MATLAB; AE 6580 Nonlinear Control

Institute of Space Technology, Islamabad, Islamabad Capital Territory PK

B.S. Aerospace Engineering, September 2018 - July 2022

- Grade: 3.93/4.00 (Summa Cum Laude)
- Valedictorian, Dean's Honor List
- Capstone Project: "Designing a Framework for Damage Tolerance Analysis of Metallic Structures
- Electives: Flight Control Systems, Spacecraft Dynamics and Controls, Guidance and Navigation of Aerospace Vehicles

HONORS AND AWARDS

PhD Scholarship, Imperial College London, 2026; UMelbourne, 2026; PoliMi, 2026; ANU, 2026

Selected for Wolfram Summer Research Institute Program, 2026
Selected for the Georgia Tech Boeing FLT Program, 2025
Fulbright Scholarship for Masters, 2024
National Award of Pakistan - International Astronomy, and Astrophysics Competition, 2024
Gold Honor - International Astronomy, and Astrophysics Competition, 2024
Achieved Academy Participant, 2024
Teknofest Turkey 2022: Winner of Black Sea Competition, Honorable Mention Prize in International Free Mission vehicle Category
Six Provisional Asteroid Discoveries - International Astronomical Search Collaboration, 2023
Institute of Space Technology: Graduated Summa Magna Cum Laude, 2022; Presidential Gold Medal for Highest grade, 2022; Vice-Chancellor's Gold Medal for best Thesis, 2022; Brigadier Dr. Atiq-ur-Rehman Graduation Prize for Best Student of the Year 2022; Valedictorian of the Year 2022; Merit Scholarship Holder, 2018-2022
Silver Honor - International Astronomy, and Astrophysics Competition, 2021
STEP Scholarship Holder for Bachelors, 2018

Computer Skills

- Languages: MATLAB, Python, C++, C#, LaTeX, SQLite, Oracle SQL, Shell Scripting, HTML, CSS, JavaScript, SysML, Julia, Wolfram, ROS, Arduino C
- Softwares: Windows, Linux Ubuntu, MS Office (Word, Powerpoint, Excel, Project, Publisher, Visio), Libre Office, ANSYS WorkBench, ABAQUS, CAD(Catia V5), Inkscape, Flight Dynamics Modelling (Athena AVL, Tornado VLM, US Datcom, XFLR5), SIMULINK, Presentation (Beamer, Lucid Press, Lucid Charts), Astrometrica, UNITY Game Development, VIM, Capella/ARCADIA (System Engineering), Valispace, Pschopy, Siemens TIA Portal
- Operating Systems: Unix/Linux, Windows, MacOS.

Academic Experience

University of Illinois Urbana-Champaign, Urbana, Illinois US
Graduate Research Assistant **Aug, 2026 -*Present***

[1] Exploitation of Reinforcement Learning Frameworks in transformers towards Hidden Markov Model Improvement.

Georgia Institute of Technology, Atlanta, Georgia US
Graduate Student Researcher **Oct, 2024 - May 2026**

[1] Explored theoretical boundaries of Nonlinear Global Model Reduction as part of the MAST Lab. (Jan, 2026 - May 2026)

[2] Researched the formulation of optimal control theory for a class of impulsively driven systems using distribution theory in the MAST Lab. (Sep, 2025 - Dec, 2025)

[3] Investigated and implemented optimal control theory on the class of positive and compartmental systems in the CASCADES Lab. (Feb, 2025 - Aug, 2025)

[4] Conducted real-time experiments on manipulator arm collaboration involving human and robotic agents for the Human-Robot Collaboration Project at the ROBOTICS Lab. (Summer 2025)

Georgia Institute of Technology, Atlanta, Georgia US
Graduate Teaching Assistant **Aug, 2025 - May 2026**

[1] Assisted in AE 6530 (Multivariable Linear Systems and Controls) and AE 6580 (Nonlinear Control).

[2] Prepared and graded assignments, and delivered technical course lectures.

Institute of Space Technology, Islamabad, Islamabad Capital Territory PK
Teaching Assistant **Feb, 2022 - June 2022**
Assisted in the preparation and delivery of lectures, and evaluated student assessments.

Peer-Reviewed Publications

[1] M. N. Rauf and E. Verriest, "Generalized Euler-Lagrange Jump Conditions for Hybrid Systems," SoutheastCon 2026, Huntsville, AL, USA, 2026, pp. 1-5, doi: 10.1109/SoutheastCon63549.2026.11476625.

[2] M. N. Rauf, E. Verriest, "Causal Infinitesimal Modeling and the Generalized Euler-Lagrange Equation in Impulsive Optimal Control", *(Under Review).*

[3] M. N. Rauf, R. A. Khan, S. I. A. Shah, and M. A. Naqvi, "Design and analysis of stability and control for a small unmanned aerial vehicle," International Journal of Dynamics and Control, Nov 2023.

[4] M. N. Rauf, R. A. Khan, and S. I. Ali Shah, "Stability analysis & development of a control strategy for a small uav," in 2023 3rd International Conference on Digital Futures and Transformative Technologies (ICoDT2), pp. 1–6, 2023.

[5] M. N. Rauf, R. Khan, and G. Majeed, "Designing an Analysis Framework for estimating life of a component subjected to a constant load and a single overload using MATLAB," Journal of Material Science & Applied Engineering, pp. 1–10, 2022.

Peer Review Service

- Reviewer, **Isospectral Steering**, *IEEE Transactions on Automatic Control* Oct, 2026
- Reviewer, **Input-to-state stability and admissibility in integral norms for linear distributed parameter systems**, *International Federation on Automatic Control IFAC* Mar, 2026

Conferences Talks

[1] "Generalized Euler-Lagrange Jump Conditions for Hybrid Systems", *SoutheastCon 2026*, Alabama, US

[2] "Stability Analysis & Development of Control Strategy for Small UAV", *International Conference on Digital Futures and Transformative Technologies*, Islamabad, Islamabad Capital Territory, PK

Professional Experience

4Earth Inc., Kennesaw, GA, United States

Automation & Controls Engineer **Jul, 2025 - Aug, 2025**
Duties: Assisted in CAD modeling and design improvements for a water treatment unit.
Programmed Siemens Programmable Logic Controllers (PLC), Siemens TIA Portal, and Human-Machine Interfaces (HMI).
Projects:

[1] Design and Implementation of an Autonomous Water Treatment unit *July, 2025 - Aug, 2025*
Suggested design and systems integration improvements, leading to a 10% increase in accessibility and data transparency.

Naqcode Technologies Pvt. Ltd., Islamabad, Islamabad Capital Territory PK

Team Lead (Flight Mechanics and Propulsion) **Sep, 2023 - Jul, 2024**
Duties: Managed technical project execution for the Propulsion and Controls teams; projects included the design and development of SRMs, UAV modeling and simulation, and the design of a UAV launch platform.
Directed the technical execution and provided oversight for flight mechanics projects, modeling framework design, and numerical simulations.
Projects:

[1] Design of Vehicle-Top UAV Launcher as well as a Catapult Launch mechanism *May, 2024 - July, 2024*
Initial Schematic Design, Structural Analysis, and Integration.

[2] Development of a Warhead system *Nov, 2023 - Mar, 2024*
Warhead System Design; Detonation Point Monitoring.

[3] Design, Performance analysis, and Optimization of SRM *Nov, 2023 - Jul, 2024*
Design and Analysis of Missile Propulsion system; Parametric studies of SRM metrics; Optimization of design variables via Genetic Algorithms.

[4] FDM and Control of Precision Glide Vehicle *Mar, 2023 - Jan, 2024*
Development of a suitable flight control law for a Precision Guided System; Structural Sizing of Control Surfaces.

[5] Design and Development of deployable SUCAV *Aug, 2022 - Jul, 2024*
Assessment of Stability; Development of Flight Dynamics Model

Flight Mechanics & Control Engineer **Aug, 2022 - Aug, 2023**
Duties: Conducted flight dynamics modeling and assessed stability and controllability for unconventional vehicles; designed missions and developed guidance and navigation strategies; designed control systems architecture, including stability augmenters and autopilots, ensuring compliance with established aerospace standards.
Projects:

[1] FDM and Control Development of Loitering Munition *Aug, 2022 - Nov, 2023*
Assessment of Stability; Development of Flight Dynamics Model; Designing and Verifying the Control Systems Architecture for 6DOF Control; Software Designing of Navigation Autopilots.

Other Experiences

GT Boeing

Mentee **May, 2025 - Sep, 2025**
Duties: Received mentorship under Hunter Kim in the GT Boeing FLT Program 2025.

Achieved Academy SGAC

EPD Team Member **Feb, 2025 - *Present***
Duties: Collaborate with the Communication team in the Education and Professional Development Sector of SGAC, led by Achieved Academy.

All Pakistan Asteroid Search Campaign, Islamabad, Islamabad Capital Territory PK

Co-Founder **Aug, 2021 - *Present***
Duties: Lead a self-launched national campaign in affiliation with the International Astronomical Search Collaboration and NASA; serve as Technical Writing Lead.
Projects:

[1] APASC Campaign 2024 *Sep, 2023 - Oct, 2023*
Scaled participation to 15 teams with 60 total participants.

[2] APASC Campaign 2023 *Sep, 2023 - Oct, 2023*
Collaborated with SGAC and Attock Astronomical Society to launch the campaign nationwide.

[3] APASC Campaign 2022 *Sep, 2022 - Oct, 2022*
Launched pilot campaign through Space Society IST in collaboration with Attock Astronomical Society.

GT CRAB Lab Robophysics Bootcamp

Participant **July, 2025 - July, 2025**
Duties: Selected as one of 10 participants for the 2025 Robophysics Bootcamp organized by CRAB

Lab. Learned principles of physics-informed robot design and constructed a 4-link snake-emulating robot.

Space Generation Advisory Council

Mentor **Dec, 2024 - May, 2025**

Duties: Participated in the 2024 Mentorship Program and served as an Academic and Professional Mentor to Carl Anthony Sta Ana.

Aviation Training Hub, Islamabad, Islamabad Capital Territory PK

Aviation Intern **Nov, 2021 - Jan, 2022**

Duties: Completed comprehensive aviation maintenance training.

Kamra Aeronautical Complex, Kamra PK

Aerospace Intern **Aug, 2021 - Sep, 2021**

Duties: Observed and studied the design and production of trainer and other aircraft.

References

Available on request

Professional Affiliations

I.	Sigma XI Society	Aug, 2026 - *Present*
II.	Techmasters (Toastmasters International) Club	Apr, 2025 - *Present*
III.	Honor Society	Feb, 2025 - *Present*
IV.	The Planetary Society	Jan, 2025 - *Present*
V.	Order of the Engineers	Dec, 2024 - *Present*
VI.	The American Society of Mechanical Engineers (ASME)	Sep, 2024 - *Present*
VII.	Institute of Electrical and Electronics Engineers (IEEE)	Sep, 2024 - *Present*
VIII.	Royal Aeronautical Society	Aug, 2024 - *Present*
IX.	Fulbright Association	May, 2024 - *Present*
X.	American Society for Testing and Materials	May, 2024 - *Present*
XI.	Institution of Engineers Pakistan	May, 2024 - *Present*
XII.	European Low Gravity Research Association	Feb, 2024 - *Present*
XIII.	Moon Village Association	Apr, 2023 - *Present*
XIV.	Space Generation Advisory Council	Feb, 2023 - *Present*

Extracurricular

GT Buzz Buddies, Atlanta, GA, USA

Mentor July, 2025 - Dec, 2025

Moon Village Association, International

Project Member of PESC project Apr, 2023 - *Present*

Space Generation Advisory Council, International

Mentor at 2024 SGAC Mentoring Program Dec, 2024 - May, 2025
Member of Space Exploration Project Group Feb, 2024 - *Present*
Project Member in Small Satellite Group Feb, 2023 - *Present*

Space Society IST (Publications Department), Institute of Space Technology, PK

Director Jan, 2022 - Sep, 2022

Google Development Student Club (IST-Chapter), Institute of Space Technology, PK

Executive Member (Publication) Sep, 2021 - Sep, 2022

Space Summer School, Institute of Space Technology, PK

Team Member (Student Management Team) July. 2022

Seventh ICASE Conference, Institute of Space Technology, PK

Team Manager (Publication) Dec. 2021